✅ 1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying significant variables

- Highest relevance.
- Students must extract from the text and table:
  - Key variables: daily production and fault rates for each product type
  - Derived variables: number of faulty units per product, total production per company, total faulty units, overall fault rate
- For example, for Electrix:
  - Faulty video players = 2,000 × 5% = 100
  - Faulty audio players = 6,000 × 3% = 180
  - Total production = 8,000; total faulty = 280 → overall fault rate = 280 / 8,000 = 3.5%
- Similarly for Tronics.
- This process exemplifies identifying core mathematical objects and their interdependencies within a complex real-world scenario—the foundational step in modeling.

✅ 2. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

- Highly relevant.
- The original context involves two companies, two product types, and differing production volumes and fault rates—making it inherently complex.
- Students must decompose it into two structurally identical subproblems (one per company), each further broken down into:

  1. Compute faulty units per product type
  2. Sum to get total faulty units and total production
  3. Calculate overall fault rate
- This "divide-and-conquer" approach is a textbook example of problem decomposition to enable mathematical treatment.

✅ 3. Competency #5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

- Highly relevant.
- Students must recognize the underlying mathematical structure:
  - Part = Whole × Percentage
  - Overall rate ≠ arithmetic mean of sub-rates, but a weighted average:

$$\text{Overall fault rate} = \frac{\sum(\text{production}_i \times \text{fault rate}_i)}{\sum \text{production}_i}$$

- A common error would be to assume Tronics' overall fault rate is (4% + 2%) / 2 = 3%, leading to an incorrect conclusion.
- Correct solution hinges on recognizing the weighted average structure, which is the mathematical essence of the problem.

✅ 4. Competencies #10 / #11: Use mathematical tools (variables, symbols, diagrams) to describe mathematical structures and relationships

- Moderately relevant.
- Although not explicitly required, effective solvers typically organize data systematically, for example:
  - Using a table:

表格

Company	Product	Production	Fault Rate	Faulty Units
Electrix	Video	2,000	5%	100
Electrix	Audio	6,000	3%	180
Total	—	8,000	—	280

  - Or writing formulas:

$$\text{Overall fault rate} = \frac{2000 \times 0.05 + 6000 \times 0.03}{2000 + 6000}$$

- This demonstrates using structured representations (tables, formulas) to express quantitative relationships.

🟡 5. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

- Weak but present relevance.
- The problem relies on implicit assumptions:
  - "Average daily" values represent typical behavior (ignoring variability)
  - Fault rates apply uniformly across all batches
  - No feedback effects (e.g., repairs don't affect next-day output)
- While students aren't asked to critique these, awareness that "this is an idealized comparison based on averages" touches on this competency.

✅ 1. 能力 #6：识别并描述现实问题中的数学要素，包括识别关键变量

> *"Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables"*

- 最高相关性。
- 学生必须从文本和表格中提取：
  - 关键变量：每类产品的日产量、故障率
  - 派生变量：每类产品的故障数量、公司总产量、公司总故障数、整体故障率
- 例如，对 Electrix：
  - 视频故障数 = 2000 × 5% = 100
  - 音频故障数 = 6000 × 3% = 180
  - 总产量 = 8000，总故障 = 280 → 整体故障率 = 280 / 8000 = 3.5%
- 对 Tronics 同理。
- 这一过程正是从复杂现实信息中识别核心数学对象及其依赖关系，是建模的起点。

✅ 2. 能力 #7：简化或分解情境，使其便于数学分析

> *"Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis"*

- 高度相关。
- 原始情境涉及两家公司、两类产品、不同产量和故障率，较为复杂。
- 学生需将其分解为两个独立但结构相同的子问题（分别计算每家公司），每个子问题又可进一步分解为：
  1. 计算每类产品故障数
  2. 求和得总故障数与总产量
  3. 计算整体故障率
- 这种“分而治之”的策略，正是通过分解使复杂问题可解的典型体现。

✅ 3. 能力 #5：识别问题中的数学结构（如规律、关系、模式）

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- 高度相关。
- 学生需识别核心数学结构：
  - 部分 = 整体 × 百分比
  - 整体比率 ≠ 各部分比率的算术平均，而是加权平均
    - 即：整体故障率 = Σ(产量$_i$ × 故障率$_i$) / Σ(产量$_i$)
- 若学生错误地认为 Tronics 的整体故障率是 (4% + 2%) / 2 = 3%，就会得出错误结论。
- 正确解法依赖于对加权平均结构的识别，这是本题的数学本质。

✅ 4. 能力 #10 / #11：使用数学工具（变量、符号、图表）描述问题中的数学结构与关系

> *"Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem"*

- 中等相关。
- 虽然题目未强制要求使用符号，但高效解题者通常会隐式或显式地组织数据，例如：
  - 列表计算：

表格

公司	产品	产量	故障率	故障数
Electrix	视频	2000	5%	100
Electrix	音频	6000	3%	180
总计	—	8000	—	280

  - 或写出公式：

$$\text{Overall fault rate} = \frac{2000 \times 0.05 + 6000 \times 0.03}{2000 + 6000}$$

- 这体现了用结构化方式（表格、公式）表达数量关系的能力。

🟡 5. 能力 #12：识别模型中的约束、假设与简化

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- 弱相关但存在。
- 题目隐含假设：
  - “日均”数据可代表典型情况（忽略波动）
  - 故障率适用于所有批次
  - 无交叉影响（如维修不影响次日生产）
- 虽未要求学生反思这些，但若能意识到“这是基于平均值的理想化比较”，则触及此项能力。